D.4.1. User Requirements and Analysis Document

Project: Canadian Labour Market Analytics: Industry Growth and Provincial Employment Trends, 2019–2025

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1. Introduction and Scope

1.1 Introduction and Purpose

The Canadian Labour Market Analytics project will examine employment growth, decline, and recovery across Canadian provinces and industries. Statistics Canada publishes monthly employment estimates by industry and geography, providing a basis for these comparisons (Statistics Canada, 2026a). The project will transform these public data into an interactive analytical report.

The solution will combine structured employment records, semi-structured metadata, and unstructured narrative text from Statistics Canada. The integrated data will be prepared, validated, stored in MySQL, and imported into Power BI for analysis and reporting.

This document defines the project's objectives, scope, stakeholder roles, functional and non-functional requirements, acceptance criteria, and revision history. It provides a shared basis for implementation, validation, and review of the completed solution.

1.2 Project Objectives

The project will enable readers to:

1. Compare employment levels across Canadian provinces and major industries.
2. Identify industries and provinces experiencing employment growth or decline.
3. Distinguish absolute employment changes from percentage changes.
4. Examine short-term fluctuations alongside longer-term employment trends.
5. Assess employment recovery relative to February 2020.
6. Explore relevant Statistics Canada commentary alongside numerical results.

The central analytical question is:

Which industries are growing in which Canadian provinces, and where has employment strengthened or recovered over time?

The project will support labour-market exploration and further investigation. Employment growth will not be interpreted as a direct measure of current vacancies or an individual's likelihood of finding employment.

1.3 Project Scope and Boundaries

In Scope

The analysis will cover January 2019 to December 2025, representing 84 reference months, across the ten Canadian provinces. Industry comparisons will use the classifications available in the selected Statistics Canada employment dataset.

Data format	Planned source	Role in the solution
Structured	CSV data from Statistics Canada Table 14-10-0355-01 (Statistics Canada, 2026a).	The project will retain monthly employment estimates for the ten provinces, the study period, and the seasonally adjusted series.
Semi-structured	JSON metadata from the Statistics Canada Web Data Service, using getCubeMetadata with productId 14100355 (Statistics Canada, 2025).	Dimension and member information will support the mapping and interpretation of geography identifiers, industry labels, and classification hierarchies.
Unstructured	Text to be scraped from The Daily Labour Force Survey releases, including the December 2025 release as an example (Statistics Canada, 2026b).	Article text will provide narrative context linked to the relevant reference month and manually reviewed province and industry tags.

The work will include source acquisition, web scraping, cleaning, transformation, integration, database implementation, analytical calculations, and interactive reporting. Raw-source snapshots, retrieval information, transformation records, and a data dictionary will support documentation and reproducibility.

Python will support web scraping and JSON parsing. Power Query will support repeatable cleaning and transformation, while Power BI will provide the analytical model, DAX measures, and interactive report (Microsoft, 2026). MySQL, a relational database management system, will store the final schema and prepared data (Oracle, n.d.-b).

The dimensional model will follow star-schema principles, with fact and dimension tables at a consistent grain (Microsoft, 2024). Power BI relationships will support filter propagation between related tables (Microsoft, 2025). MySQL foreign key constraints will support referential integrity in the database (Oracle, n.d.-a).

Out of Scope

- The three Canadian territories and analysis below the provincial level.
- Employment forecasts beyond December 2025.
- Individual job advertisements, applicants, salaries, and personal employment data.
- Real-time reporting and automatically scheduled updates.
- Causal conclusions about the drivers of employment changes.

Historical articles will provide context for interpretation. Differences between figures in these articles and the retrieved employment estimates will be documented where identified.

1.4 Stakeholder Roles

The report will serve a broad readership with different analytical interests. The project team will operate the preparation workflow and maintain the solution, while report users will explore the analytical results.

Stakeholder	Role and responsibility
Students, recent graduates, and job seekers	Explore provincial and industry employment patterns to support further career research.
Career advisors and educational institutions	Use regional and sector comparisons to inform guidance and discussions about labour-market trends.
Employers and recruitment agencies	Examine employment trends to identify industries and regions for further investigation.
Business analysts, researchers, and economic-development organizations	Compare employment levels, growth, decline, and recovery across provinces and industries.
Project Coordinator and Data Preparation Analyst	Coordinate the work; acquire, scrape, clean, and integrate data; maintain source records and transformation documentation.
Data Model and DAX Analyst	Create and populate the MySQL schema, validate relationships, and develop analytical measures.
Power BI Visualization and Dashboard Analyst	Connect Power BI to MySQL, design report pages, test visual interactions, and prepare supporting documentation.

Instructor or teaching assistant	Review the project's alignment with course requirements and provide feedback.

Report users will select filters, compare visualizations, and inspect supporting context. Source preparation, database maintenance, and changes to analytical calculations will remain the responsibility of the project team.

2. Functional Requirements

The functional requirements describe the behaviours and capabilities of the complete solution, including the preparation workflow operated by the project team and the Power BI report used by readers. Each requirement has a unique identifier to support implementation tracking and linkage to acceptance criteria.

High-priority requirements will be implemented first because they support the core workflow, analytical objectives, and mandatory course requirements. Medium-priority requirements will follow the core workflow and support detailed exploration and interaction. Both priorities remain within the planned final scope.

2.1 Prioritized Functional Requirements

ID	Priority	Requirement
FR-01	High	The system shall import the CSV source specified in Section 1.3 and retain only records with Statistics set to Estimate and Data type set to Seasonally adjusted, for the ten Canadian provinces and reference months from January 2019 to December 2025.
FR-02	High	The system shall parse the JSON metadata, flatten geography and industry member information, and map source identifiers to labels. Unmatched categories shall be recorded for review.
FR-03	High	The acquisition workflow shall scrape article titles, publication dates, reference months, article text, source URLs, and retrieval timestamps. It shall retain raw HTML snapshots, remove navigation and boilerplate text, identify duplicate URLs, and log failed retrievals.
FR-04	High	The preparation workflow shall standardize dates and numeric fields, separate NAICS codes from industry labels where applicable, and identify duplicate month–province–industry records. It shall preserve units and source flags and retain suppressed or unavailable values as nulls.

FR-05	High	The system shall integrate employment records with metadata and associate articles with reference months and manually reviewed province and industry tags. Article records and bridge tables shall remain separate from employment facts so that multiple articles do not multiply employment values.
FR-06	High	The system shall create and populate Date, Province, and Industry dimensions, an Employment fact table, an Article table, and article tag bridge tables in MySQL. It shall enforce primary and foreign keys and make the prepared tables available for import into Power BI.
FR-07	High	The report shall provide province, industry, and date-range filters and a reference-month selection for employment-level comparisons. Relevant visualizations shall respond consistently to the selected filters.
FR-08	High	The report shall display the employment level for the selected reference month in thousands of persons. It shall not calculate a selected-period employment level by summing monthly employment estimates.
FR-09	High	The report shall calculate absolute and percentage month-over-month and year-over-year changes using the comparison and missing-value rules in Section 2.2.
FR-10	High	The report shall calculate a trailing 12-month average using the selected month and the preceding 11 calendar months. The result shall be unavailable if any required monthly estimate is missing or suppressed.
FR-11	High	The report shall calculate an employment recovery index with February 2020 equal to 100 and identify sustained recovery using Section 2.2. It shall distinguish no observed decline, no recovery observed, and insufficient data.
FR-12	High	The report shall provide an industry-by-province matrix showing employment levels or percentage changes with conditional formatting for growth and decline, and a line chart showing monthly employment and the 12-month rolling average.
FR-13	Medium	The report shall allow users to search the available article text and view relevant excerpts and source links. Articles shall be associated with reference months and reviewed tags, and the report shall indicate when no supporting article is available.
FR-14	Medium	The report shall provide an industry drill-through page showing employment levels, changes, the rolling average, recovery measures, provincial comparisons, and historical trends.
FR-15	Medium	The report shall provide a bookmark toggle between absolute employment and year-over-year percentage-change views. Titles and labels shall identify the displayed measure, unit, and comparison period.

2.2 Analytical and Data Interpretation Rules

Employment Units and Aggregation

The source table reports employment estimates in thousands of persons (Statistics Canada, 2026a). The report will retain this scale and label it consistently. Missing or suppressed estimates will remain unavailable and will not be replaced with zero.

The project will define one employment fact record per month, province, and industry for the selected series. Aggregation will use compatible, non-overlapping categories. Parent industry totals will not be added to their component industries.

Month over Month and Year over Year Changes

Month-over-month change will compare the selected month with the immediately preceding calendar month. Year-over-year change will compare the selected month with the same month one year earlier.

Absolute change will equal the current employment estimate minus the comparison estimate. Percentage change will equal this difference divided by the comparison estimate, multiplied by 100.

The result will be unavailable when a required estimate is missing. A percentage change will also be unavailable when the comparison estimate is zero. These cases will be distinguished from a valid zero change.

Twelve-Month Rolling Average

The rolling average will use the 12 consecutive calendar months ending with the selected month. All 12 monthly estimates must be valid. Missing months will not be skipped or replaced with zero.

Recovery Measurement

For this project, the team defines sustained recovery as the first month of the earliest sequence of three consecutive valid months at or above the February 2020 employment level, following an observed post-February 2020 decline below that level. This is a project-specific operational definition rather than an official Statistics Canada recovery classification.

The recovery index will equal the selected month's employment estimate divided by the February 2020 estimate, multiplied by 100. The index will be unavailable when the baseline is missing or zero.

A missing month will interrupt a qualifying three-month sequence. Series with no observed decline, no recovery observed, and insufficient data will be reported separately. Gaps that prevent the classification will not be treated as evidence of no decline or no recovery.

Article Dates and Source Context

Publication dates and reference months will be stored separately. For example, the Labour Force Survey release for December 2025 was published on January 9, 2026 (Statistics Canada, 2026b). Articles will therefore be linked to the period they discuss, rather than to their publication month alone.

Article text will support interpretation and will not overwrite the numerical employment estimates. Any differences between figures quoted in historical articles and the retrieved CSV estimates will be documented.

3. Non-Functional Requirements

Non-functional requirements specify the quality standards for the data preparation workflow, MySQL database, and Power BI report. The following thresholds are proposed project targets for testing and approval; they are not claims that the solution has already met them.

ID	Attribute	Requirement	Verification / acceptance threshold
NFR-01	Data accuracy	Validated employment estimates shall retain their source value, reference month, geography, industry, seasonal-adjustment status, and unit (thousands of persons).	For a reproducible sample of at least 30 imported fact rows, 100% agree with the corresponding Statistics Canada source records; any discrepancies are logged.
NFR-02	Data integrity	The database shall prevent duplicate fact-grain records and enforce referential integrity across employment, dimension, article, and bridge tables.	SQL checks find zero duplicate selected-series month–province–industry keys and zero orphan foreign keys.
NFR-03	Performance	The report shall respond promptly to common report navigation and filter interactions in its documented test environment.	Across 10 repeat runs using the agreed test computer and dataset, each core report page loads within 5 seconds and standard filter interactions complete within 3 seconds in at least 9 runs.

NFR-04	Access security	Only authorized project team members shall modify the MySQL schema or prepared data; report readers shall not require database write privileges.	A read-only test account cannot INSERT, UPDATE, DELETE, or ALTER prepared tables; authorized account permissions are documented.
NFR-05	Reliability and error handling	Failed source retrievals, unmapped metadata, and invalid or unavailable values shall be logged rather than silently dropped, replaced by zero, or guessed.	A controlled test with an invalid URL and a missing value produces explicit error/log records; the missing value remains null.
NFR-06	Usability	Report controls, titles, legends, measure labels, and missing-data states shall be unambiguous to a new reader.	Three reviewers' complete province/industry filtering, a year-over-year comparison, and drill-through unaided; each identifies the displayed unit and comparison period correctly.
NFR-07	Maintainability	Data preparation, MySQL objects, DAX measures, and source-to-report mappings shall be reproducible and documented.	A project-team reviewer can follow the saved runbook to rebuild prepared tables and refresh the report with matching validation totals.
NFR-08	Source traceability	Acquired CSV, JSON, and article inputs shall have source identifiers and retrieval details; transformations shall be recorded.	Every prepared source family has its URL or identifier, retrieval date, raw snapshot or preserved input, and transformation log.
NFR-09	Source-use compliance	The solution shall attribute Statistics Canada datasets and articles and follow applicable public-source conditions; no personal employment records shall be collected.	A source attribution checklist is complete, original article URLs are available, and schema inspection shows no individual job-seeker records.
NFR-10	Compatibility	Prepared MySQL tables and relationships shall be consumable by the intended Power BI model without loss of data type, scale, or grain.	A complete import/refresh succeeds, date and numeric columns have corrected types, and independently checked aggregated values match SQL results.

4. Acceptance Criteria

An acceptance criterion is a pass/fail condition linked to a requirement from Section 2. Evidence may include sample-row comparisons, SQL query outputs, screenshots, calculation worksheets,

test logs, and report demonstrations. All criteria below are intended for verification upon implementation; no tests are represented as already completed.

AC ID	Linked FR	Test focus	Pass criterion	Suggested evidence
AC-01	FR-01	Import and scope	The imported data contain only records with Statistics set to Estimate and Data type set to Seasonally adjusted, for the ten Canadian provinces and reference months January 2019–December 2025. Source-to-target row counts are reconciled, and coverage across the 84 reference months is documented.	Import log, scope query, source comparison
AC-02	FR-02	Metadata mapping	All included geography and industry identifiers have readable mapped labels, or each unresolved identifier appears in a documented exception log; sample mappings agree with source metadata.	Mapping tables and unmatched-value log
AC-03	FR-03	Article scraping	At least one Labour Force Survey release page shall be successfully scraped from a documented list of target URLs. Each retrieved article shall include its title, publication date, reference month, body text, source URL, retrieval timestamp, and raw HTML snapshot. Duplicate URLs and failed retrievals shall be logged. The acquisition log shall report the numbers of target, successful, and failed pages.	Article sample, saved HTML, scraper log
AC-04	FR-04	Cleaning	Reference periods are standardized, fields have the correct data types, separable NAICS codes are captured, duplicate fact keys are absent, and unavailable or suppressed entries remain null with flags retained.	Data-quality SQL tests, transformation steps
AC-05	FR-05	Integration	Articles are related to correct reference months and reviewed tags; joining through separate article/bridge structures does not increase the employment fact count or inflate total employment.	Before/after row-count and aggregate checks
AC-06	FR-06	Database model	Date, Province, Industry, Employment, Article, and article-tag bridge tables exist; configured keys and relationships pass integrity tests; Power BI imports the prepared data successfully.	Schema diagram, SQL tests, refresh screenshot

AC-07	FR-07	Filters	Province, industry, date range, and reference-month selection restrict applicable visuals correctly. A fixed test selection returns the same figures as the corresponding SQL query.	Power BI screenshot and comparison query
AC-08	FR-08	Employment level	Selected-month employment values show thousands of persons. A multi-month selection never adds monthly employment estimates together and labels distinguish the selected reference month.	Measure test and visual screenshot
AC-09	FR-09	Change measures	MoM and YoY absolute and percentage changes match independently computed examples. Missing comparison values return blank; zero denominators return blank for percentages; a valid zero change remains zero.	Worked calculation checks and edge-case tests
AC-10	FR-10	Rolling average	For a complete series, the trailing average equals the arithmetic mean of the selected month plus the preceding 11 calendar months. Any missing/suppressed month makes the value unavailable.	12-month comparison worksheet and missing-month test
AC-11	FR-11	Recovery index	February 2020 is 100 where a valid baseline exists. Classification identifies the first of three consecutive qualifying months after an observed decline; cases with no observed decline, no observed recovery, or insufficient data are distinguished.	Four recovery scenario checks
AC-12	FR-12	Core visuals	Industry-by-province matrix shows the chosen employment measure with understandable growth/decline formatting; line chart shows both monthly estimates and valid 12-month averages, matching underlying measures.	Visual inspection and source-value checks
AC-13	FR-13	Article context	Searching available text returns matching articles, excerpts, and working source links; month and reviewed tags determine relevance; an empty result displays a clear no-article message.	Search, link, and empty-state demonstrations
AC-14	FR-14	Industry drill-through	Drill-through carries the selected industry into its detail page and displays consistent province comparison, history, rolling average, changes, and recovery measures.	Drill-through screenshot and filter check

AC-15	FR-15	Bookmark toggle	Switching between absolute employment and YoY percentage change updates the measure, labels, units, and comparison-period wording without inconsistent or stale titles.	Before/after screenshot and selected-value checks

4.1 Final Acceptance and Sign-off

The implemented solution will be ready for final academic review when AC-01 to AC-15 and the verification criteria for NFR-01 to NFR-10 have passed, or any exceptions have been documented and approved by the appropriate reviewer. High and medium priorities determine implementation order. Any proposed change to mandatory course requirements requires approval from the instructor or GA. Test evidence, outstanding issues, and review decisions shall be recorded. Instructor approval shall be recorded only when received.

Review field	**Record**
Functional tests passed	Not yet tested
Non-functional checks passed	Not yet tested
Outstanding issues / accepted deviations	To be recorded during testing and review
Reviewed by	Pending review
Review date	Pending review
Final status	Pending

5. Revision History

The revision history records completed changes to this document, including the version, revision date, description of changes, responsible team member or team, and review status.

Version	**Date**	**Change / scope**	**Prepared or reviewed by**	**Status**
0.1	October 19, 2026	Initial introduction, objectives, scope, and stakeholder roles (Section 1).	Nguyen Van Sa	Draft prepared
0.2	October 20, 2026	Prioritized functional requirements and analytical/data interpretation rules (Section 2).	Nguyen Van Sa	Draft prepared

0.3	October 21, 2026	Draft Sections 3–5: quality requirements, FR-linked acceptance criteria, and controlled revision history.	Raviranjan Rajput; Mujeeb Ahmed Sayed	Draft prepared
1.0	October 22, 2026	Consolidated requirements document following team review and requested revisions.	Group 3	Approved
1.0	To confirm	Pending instructor review and approval	Instructor	Not yet approved

References

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